

External Reviewer Packet

Adam Okulicz-Kozaryn

Tuesday 28th April, 2026 14:09

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1 statements

2 cv

ADAM OKULICZ-KOZARYN [updated: 2026:4]

Department of Public Policy & Administration
Rutgers University, The State University of NJ
401 Cooper St, Camden, NJ 08102

adam.okulicz.kozaryn@gmail.com
<https://theaok.github.io>
<http://scholar.google.com/citations?hl=en&user=pz9RYIoAAAAJ>

EMPLOYMENT

Associate Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2018-??

Assistant Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2012-18

Research Scientist at Texas Schools Project (2010-11) and Clinical Assistant Professor, UT-Dallas, 2010-12

EDUCATION

Post-doctoral Fellow, Institute for Quantitative Social Science, Harvard University, 2008-10

PhD Public Policy and Political Economy, The University of Texas at Dallas, 2008

MS Corporate and Economic Policy, Poznan University of Economics (Poland), 2003

BS Banking and Finance, Poznan School of Banking (Poland), 2001

MAJOR PUBLICATIONS (all on Google Scholar; publications with students listed under service; AOK=Adam Okulicz-Kozaryn; click text for PDF)

2026 Golden L, Valente RR, AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research

Urbanization and Happiness. A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally. Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing

2025 AOK, Martinez L Colombia: Unlivable but Happy. Fool's Paradise? (No, a Real Paradise, Better than the US) Applied Research in Quality of Life

2024 D'Italia M, AOK, Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS) Social Indicators Research

Urban Regrets (Unhappy Metros: Satisfaction With Life Scale (SWLS)) Heliyon

2023 Magdy Zaki E, AOK, Valente RR, Subjective Well-being and Urbanization in Egypt Cities

AOK, Everett B, Magdy Zaki E, Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones) Urban Affairs Review

2022 Unhappy Metros: Panel Evidence Applied Research in Quality of Life

AOK, Valente R, Misanthropolis: Do Cities Promote Misanthropy? Cities

2021 AOK, Valente R, Urban unhappiness is common Cities

AOK, Morawski L, A similar effect of volunteering and pensions on subjective wellbeing of elderly Economics and Sociology

2020 Valente R, AOK, Religiosity and Trust: Evidence from the United States Review of Religious Research

AOK, Valente R, The Perennial Dissatisfaction of Urban Upbringing Cities

AOK, Morawski L, Effect of volunteering and pensions on subjective wellbeing of elderly—are there cross-country differences?
Applied Research in Quality of Life

2019 Are we happier among our own race? Economics and Sociology

AOK, Altman M, The Energy Paradox: Energy Use And Happiness Applied Research in Quality of Life

2018 AOK, Valente R, No Urban Malaise for Millennials Regional Studies

2017 AOK, Valente R, Livability and Subjective Wellbeing Across European Cities Applied Research in Quality of Life

Strzelecka M, AOK, Is tourism conducive to residents' social trust? Evidence from large-scale surveys. Tourism Review

AOK, Valente R, Life Satisfaction of Career Women and Housewives Applied Research in Quality of Life

AOK, Valente R, City Life: Glorification, Desire, and the Unconscious Size Fetish, ed. by I Kapoor, in Psychoanalysis and the Global, University of Nebraska Press

AOK, Golden L, Happiness is flextime Applied Research in Quality of Life

2016 AOK, Strzelecka M, Happy Tourists, Unhappy Locals Social Indicators Research

Happiness Research for Public Policy and Administration Transforming Government: People, Process and Policy

Unhappy Metropolis (When American City Is Too Big) Cities

AOK, Mazelis J M, More Unequal in Income, More Unequal in WellBeing Social Indicators Research

AOK, Mazelis J M, Urbanism and Happiness: A Test of Wirth's Theory of Urban Life Urban Studies

2015 Freedom and Life Satisfaction in Transition Society and Economy

AOK, Nash T, Tursi N O, Luxury Car Owners Are Not Happier Than Frugal Car Owners International Review of Economics

The More Religiosity, the Less Creativity Across US Counties Business Creativity & the Creative Economy

Income Inequality and Wellbeing Applied Research in Quality of Life

! !: this book is excluded from the promotion packet (this is a popular science book) Happiness and Place: Cities v Nature. Why Life is Better Outside of the City? Palgrave Macmillan

2014 AOK, Holmes IV O, Avery D R, The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals
Journal of Applied Psychology

Winners and Losers in Transition: Preferences For Redistribution and Nostalgia For Communism In Eastern Europe
Kyklos

'Freedom From' and 'Freedom To' across countries Social Indicators Research

Natural Sprawl Administration & Society (Disputatio Sine Fine section) (section of the journal reviewed by editor only)

2013 Cluttered Writing: Adjectives and Adverbs in Academia Scientometrics

City Life: Rankings (Livability) vs Perceptions (Satisfaction) Social Indicators Research

2012 Does Religious Diversity Make Us Unhappy? Mental Health, Religion & Culture

Income and Well-being across European Provinces Social Indicators Research

2011 Berry B J L, AOK, An Urban-Rural Happiness Gradient Urban Geography

Geography of European Life Satisfaction Social Indicators Research

Europeans Work To Live and Americans Live To Work (Who is Happy to Work More: Americans or Europeans?)
Journal of Happiness Studies

2010 Religiosity and Life Satisfaction Across Nations Mental Health, Religion & Culture

2009 Berry B J L, AOK, Dissatisfaction with City Life: A New Look at Some Old Questions Cities

EXTERNAL GRANTS

190k PLN (~\$50k) PI, "The effect of social transfers and social capital on happiness of elderly." Vistula University, Warsaw Poland, Polish National Science Foundation (Narodowe Centrum Nauki NCN) 2016/21/B/HS4/03058 .. 2017-2019

\$1.5k Consultant (summer salary), "Health information inequity in South Jersey-Vineland" South Jersey Institute for Population Health (SJIPH) Catalyst Grant <https://sjiph.org/projects/health-information-inequity-in-south-jersey-vineland> 2026

INTERNAL GRANTS (AWARDS and FELLOWSHIPS)

\$12k faculty advisor, Rutgers Camden chancellor's dissertation completion award for Marlo Rossi2026

\$30k faculty advisor, Rutgers Camden urban innovation fund grant to Ojobo Agbo Eje2024-2026

\$15k PI, Rutgers Camden chancellor's interdisciplinary research collaboration grant 2023-2024

\$1k Rutgers-Camden Research Council Grant Award 2016-2017

\$1k Rutgers-Camden Digital Teaching Fellow 2016-2017

\$4k Rutgers-Camden Rand Institute Faculty Fellow 2013-2014

\$1k Rutgers-Camden Civic Engagement Faculty Fellow 2013-2014

TEACHING (note curriculum development: [developed/designed] or [re-designed])

2025 recipient of department's Jay A Sigler award for "outstanding service and dedication to students" (awarded by students); the plaque says "this award is given to the faculty member who exhibits a commitment to student's academic development and personal growth."

GIS for data science <https://theaok.github.io/gisPy/> [developed/designed] 2023-present

Data Processing (data management for data science) <https://theaok.github.io/datManPy> [developed/designed] 2023-present

Data Visualization <https://theaok.github.io/vis> [developed/designed] 2023-present

Coping with Extremes: Ecological resilience & human well-being [innovative class co-taught with biology dept] [co-developed/designed]	2019
Community & WellBeing (Happiness and Place) [developed/designed]	2018-present
Data Management [developed/designed]	2016-present
GIS for Public Sector/Urban mapping [developed/designed]	2012-present
Quantitative Methods 2 [re-designed]	2013-2017, 2024
Research Methods [re-designed]	2012-present
Introduction to Regional and Economic Development [re-designed]	2012-2014
Quantitative Methods 1 [re-designed]	2012-2013

SERVICE

SERVICE TO THE FIELD/PROFESSION:

Review papers (approx 3-4/mo or 36-48/year as of Jan 2026; “top journals” underlined; starting in 2022: last 2 digits of year of review shown):

- <informal/for a colleague> [24]
- Acta Sociologica
- Applied Economics [23,23]
- Applied Research in Quality of Life [23,23,23,24,24,24,24,26]
- Basic and Applied Social Psychology [22]
- BMC Psychology [26]
- BMJ Open
- British Journal of Political Science [25]
- Cities [22,23,23,23,23,23,24,24,24,24,24,25,25,25,25,25,25,25,25]
- City, Culture and Society
- City, Territory and Architecture [24]
- Cogent Arts & Humanities [23]
- Cognition and Emotion
- Communications Psychology [23]
- Comparative Economic Studies [26]
- Discover Public Health [25,26]
- Ecological Economics [24]
- Environment and Planning A
- Economics and Politics
- European Journal of Social Psychology
- European Journal of Political Economy [23,23,24]
- European Societies
- European Urban and Regional Studies [22,22]
- Frontiers in Psychology [23]
- Global Transitions [24]

- Humanities and Social Sciences Communications [23,23,23,24,24,25,25]
- Heliyon [23,24]
- International Journal of Comparative Sociology [23,24,24,25]
- International Journal of Environmental Research and Public Health
- International Journal of Psychology
- International Journal of Social Economics [22]
- International Journal of Wellbeing [25]
- International Perspectives in Psychology: Research, Practice, Consultation
- International Sociology
- Journal of Comparative Economics
- Journal of Contemporary Religion
- Journal of Economic Growth
- Journal of Economic Psychology
- Journal of Economic and Social Geography
- Journal of Personality and Social Psychology: Personality Processes and Individual Differences[16]
- Journal of Personality and Social Psychology: Attitudes and Social Cognition[26]
- Journal of Religion and Health
- Journal of Regional Science
- Journal of Happiness Studies [22,23,24,26,26,26]
- Journal of Hospitality and Tourism Management
- Journal of Public and Nonprofit Affairs
- Journal of Public Policy [23,23]
- Journal of the American Planning Association
- Journal of Urbanism
- Land Use Policy [23,24,24]
- Local Development and Society [25]
- Mental Health, Religion and Culture
- Millah: Journal of Religious Studies [25]
- Nature Communications [24]
- Open Psychology Journal
- Quality of Life Research
- Papers in Regional Science [22,22]
- PLOS ONE [22]
- Political Behavior [22]
- PNAS Nexus [24]
- Problems of Post-Communism [22]
- Psychology of Religion and Spirituality
- Public Administration Review [24,25]
- Regional Studies [22,22]
- Review of Political Economy
- Routledge (book publisher) [22]
- Rural Sociology [22]
- Urban Geography
- Urban Studies [25]

- Urban Affairs Review
- Utilities Policy [25,25]
- Science of the Total Environment
- Scientific Reports [24]
- Scienze Regionali Italian Journal of Regional Science
- Scientometrics [22,22]
- Semestre Economico (Universidad de Medellín) [22]
- Social Forces
- Social Psychological and Personality Science
- Social Indicators Research [as reviewer] [22,22,23,23,23,23,24,24,24,24,25,25,25,25,25]
- Social Indicators Research [as co-editor]¹ [25, 25, 25, 25, 25]
- Social Science And Medicine [24,24,25,25,25,25,25]
- Social Science Research [23]
- Social Sciences & Humanities Open [23]
- Sociological Spectrum
- Sociology of Religion
- Sustainability
- Sustainable Cities and Society
- Sustainable Futures [25]
- The Lancet Planetary Health (IF >10)
- Third World Quarterly [22,23]
- Wellbeing, Space & Society [22,25]

Reviewed books chapters:

- 3 chapters in “A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally.” Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing [26,26,26]

Reviewed books for:

- Routledge [22]
- Cambridge U Pr [23]

Served on editorial boards:

- Social Indicators Research: Associate Editor / Co-Editor² Jan 2024-??
- Political Behavior Jan-Dec 2019

Reviewed grant proposals:

- Polish National Science Foundation (NCN)
- Israel Science Foundation (ISF)

SERVICE TO STUDENTS:

Publications with students (all students unless indicated: “[not student]”):

¹As a co-editor of Social Indicators Research, conducted through reviews (printed out, marked up, etc). Note: this is different from regular editing, where I just skim through the paper or read some parts only; and hence this extra work is listed here separately from editing as reviewing [starting oct2025 each such review marked with 2 last digits of year]

²This is not just sitting on the board of the journal and having an occasional board meeting, I am editing papers every week spending as of 2026 $1 + hrs/wk$ (earlier as i was less experienced it was $2 + hrs/wk$) on skimming through the submitted manuscripts, inviting reviewers, reading reviews, and making editorial decisions.

10. Golden L [not student], Valente RR [not student], AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research 2026
9. AOK, L Martinez [not student], E Mikhaeil; Happy Colombia, Unhappy Bogota (urban-rural happiness gradient in Colombia) Wellbeing, Space and Society 2026
8. E Mikhaeil, AOK; Well-Being and the authoritarian welfare state: sector of employment and human happiness in Arab countries; Public Organization Review 2025
7. MJ D'Italia, AOK; Constructing General Human Agency Indicators (GHAls) and a General Personal Agency Scale (GPAS); Social Indicators Research 2025
6. E Mikhaeil, AOK; Public-private job satisfaction differential: The case of Egypt; Public Organization Review 2025
5. AOK, BK Everett, E Mikhaeil; Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones); Urban Affairs Review 2024
4. E Mikhaeil, AOK; Subjective well-being and urbanization in Egypt; Cities 2024
3. S Deb, AOK; Exploring the association of urbanization and subjective well-being in India; Cities 2023
2. FA Abarghuie, R Javadi, E Mirabi, AOK; Experience of urban thermal pleasure: A qualitative study to the understanding of thermal-scape quality; Cities 2022
1. R Stansfield [not student], E Aaronson, AOK; Police use of firearms: Exploring citizen, officer, and incident characteristics in a statewide sample; Journal of criminal justice 2021

Supervised dissertations for [year of graduation, first job]:

9. Isabelle Surielow [starting dissertation hours in Fall 2026]
8. Wei Chen [28??]
7. Erick W Udogu [27??]
6. Douglas N Zacher [27??]
5. Marlo Rossi [26??]
4. Michael D'Italia [24, RU-Camden] PDF
3. Shibin Yan [24, Walter Rand Institute for Public Affairs] PDF
2. Aditi Manke [21, Virginia Tech] PDF
1. Shourjya Deb [20, SIPRI] PDF

Supervised dissertations for less than the whole duration:

1. Helena Cabezas [Fall 25 only]

Served on dissertation committees:

10. Jason Schweitzer (Prevention Science) [27??]
9. Daniel Assamah [27??]
8. Monty Sanders [26??]
7. Brian Everett [27??]
6. Kate Cruz [22]
5. Charles Ian Hugh Forsyth (University of Otago NZ) [22]
4. Straso Jovanovski [19]
3. Spencer Clayton [18]
2. Rasheda Weaver [17]
1. Chris Wheeler [17]

Served on dissertation committees for less than the whole duration:

1. Yanan Li [stopped serving just before the proposal in 2023]

Supervised data science master thesis for [year of graduation]:

6. Nalluri Chandrahas [26]
5. Bhavyani Dodda [25]
4. Thanvitha Reddy Munagala [25]
3. Pavan Satya Venigalla [25]
2. Het Rakeshbhai Patel [25]
1. Raman Srinivas Naik [25]

Served on master thesis committees in social science:

2. Ethan Aaronson (crim) [20]
1. Rachel Wagner (psy) [20]

Independent/Directed Study [starting in 2023 counts shown (2014-present)]:

- fa25 x1
- sp25 x4
- fa24 x7
- sp24 x2
- fa23 x1

Researchers/Predocs/Postdocs/Lab Members/Visiting Scholars:

- hired 2 data scientists (Sep 2025-Jul 2026)
- planning to host Visiting Scholar, Professor Leonie C. Steckermeier Fall 2026

DEPARTMENTAL AND CAMPUS-WIDE SERVICE:

At Rutgers-Camden served on the following departmental committees:

- Awards/Events, chair 2013-16, 2021-22; 2023-25
- Awards/Events, member 2012-13, 2019-20, 2023
- Strategic Planning, member 2019-20
- PR/Website, chair 2014-16
- PR/Website, member 2012-13
- PhD, member 2014-21
- Methods Qualifying Exam, member 2012-21

Other Rutgers-Camden departmental service:

- held dissertation seminars for a group of doctoral candidates (in addition to one-on-one) Fall 2025
- led Brown Bag Seminars 2018-19; 2023-24
- As co-editor of a journal organized "Publish or Perish" workshop Oct 2024

At Rutgers-Camden served on the following campus-wide committees:

- AI committee Fall 2025
- Advancement and Promotions (A&P), member/ad hoc chair 2020-25
- Advancement and Promotions (A&P), member 2018-19
- Faculty Senate 2018-25
- Faculty Life, chair 2019-22
- Faculty Life, member 2014-15

Other Rutgers-Camden campus-wide service:

- led creation and maintenance of data science lab and MS in data science and associated certificate programs 2018-22

- served as director for data science certificate: <https://graduateschool.camden.rutgers.edu/graduate-programs/#certificates> 2022-??
- organized Geographic Information System (GIS) workshop attended by faculty from across the campus <https://theaok.github.io/shortGIS/> 2017

COMMUNITY SERVICE

provided data management for:

- Ohio Business Round Table (contracted by Miami University, Ohio; Contract Amount: \$7,000) 2022
- 2018 Michigan Economic Competitiveness Study, Michigan Chamber of Commerce (contracted by Northwood University, Michigan; Contract Amount: \$5,000) 2018

assisted following Rutgers-Camden students in their community projects as a faculty advisor:

- data science MS Ojobo Agbo Eje: "Camden Health Equity Data Visualization Initiative" project: [click for news item] [hired/supervised 4 students, budget \$30k] 2024-26
- public affairs PhD Zach Chengcheng Yue at Walter Rand Institute: Covid-19 in Atlantic County ... Spring-Fall 2025
- public affairs PhD Marlo Rossi at Walter Rand Institute: Abortion in South Jersey; and other projects on abortion: [click for news item] Fall 2025-Spring 2026
- MPA Olorunfunmi Adebajo's internship at the Camden County Community Development Office's Homelessness Assistance Unit: [click for news item], 2024

other community service:

- made GIS maps dashboard (containing work by students in my GIS class): available on campus or via RU VPN only: <https://maps-dppa.camden.rutgers.edu/index.php?search=map&title=Special%3ASearch&profile=default&fulltext=1> 2019-23
- assisted Hunterdon Central Regional High School student: happiness paper class project Fall 2025-Spring 2026
- organized Geographic Information System (GIS) workshop for The Walter Rand Institute for Public Affairs <https://theaok.github.io/shortGIS/vShortGeoda.pdf> 2019

MOST FREQUENT RECENT CONFERENCES

LASA (Latin American Studies Association) 2024

AAG (American Association of Geographers): 2015, 2016, 2017, covid19/none,

ISQOLS (International Society for Quality of Life Studies): 2015, 2017, 2018, covid19/none, 2023, 2024 (webinar), 2026 (regional)

ERSA (European Regional Science Association): 2015, 2017, 2018, covid19/none, 2023

AFFILIATE SCHOLAR

- Australian Centre on Quality of Life (2016-present)
- CURE: Center for Urban Research and Education, Rutgers-Camden (2012-18)
- EHERO: Erasmus Happiness Economics Research Organization, Erasmus, the Netherlands (2013-16)
- Vistula University, Warsaw, Poland (2015-18)

3 papers: various social indicators

- 3.1 The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being (Okulicz-Kozaryn and Altman 2019)



The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being

Adam Okulicz-Kozaryn¹ · Micah Altman²

Received: 3 April 2018 / Accepted: 27 February 2019 / Published online: 12 March 2019
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Abstract

Earth's per capita energy use continues to grow, despite technological advances and widespread calls for reduction in energy consumption. The negative environmental consequences are well known: resource depletion, pollution, and global warming. However many remain reluctant to cut energy consumption because of the widespread, although, implicit, belief that a nation's well being depends on its energy consumption. This article systematically examines the evidential support for the relationship between energy use and subjective well-being at the societal level, by integrating data from multiple sources, collected at multiple levels of government, and spanning four decades. This analysis reveals, surprisingly, that the most common measure of subjective well-being, life satisfaction, is unrelated to energy use – whether measured at the national, state or county level. The nil relationship between happiness and energy use is reminiscent of the well-known Easterlin Paradox, however the causal mechanisms responsible to each remain in question. We discuss the possible causes for the Happiness-Energy paradox and potential policy implications.

Keywords Energy use · Energy consumption · Energy intensity of economy · Sustainability · Happiness · Life satisfaction · Subjective well-being (SWB)

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s11482-019-09719-y>) contains supplementary material, which is available to authorized users.

✉ Adam Okulicz-Kozaryn
adam.okulicz.kozaryn@gmail.com

Micah Altman
micah.altman@gmail.com

¹ Department of Public Policy, Rutgers, The State University of New Jersey, Camden Campus 401 Cooper St, Camden, NJ 08102, USA

² Massachusetts Institute of Technology, Cambridge, MA, USA

Introduction

The environmental consequences of human energy consumption are one of the world's biggest problems. In particular, energy consumption is a key driver of climate change (MacKay 2008; Pachauri et al. 2014). For instance, 98% of US carbon dioxide emissions are due to energy consumption (Energy Information Administration 2008). Despite technological advances, Earth's per-capita energy use has increased about 40% over the past four decades, and continues to grow. An essential policy and ethical challenge is to find ways to be environmentally efficient at production of human well-being, to achieve good life for all within planetary boundaries (Dietz et al. 2009; O'Neill et al. 2018).

Energy is a strategic resource. Countries wage wars over energy sources, and much of politics is driven by energy. Many countries rely heavily on energy production and many use it as a political tool. Virtually all countries seek to obtain more energy sources. A recent example is fracking (Inman 2014). Yet, we need to consume less—for at least two obvious reasons: First, worldwide, most of the energy consumed is non-renewable, and this pattern of use will persist for some time (MacKay 2008). Second, energy consumption causes pollution, and pollution harms the environment, other species, and ourselves (MacKerron and Mourato 2009; Gandelman et al. 2012; Ferreira et al. 2013). Thus, natural resource depletion and pollution potentially cancel out the benefits of energy consumption.

Many argue that consumption must be reduced dramatically. A recent report by an Intergovernmental Panel on Climate Change is alarming (Pachauri et al. 2014 and <http://www.ipcc.ch>).

A major review estimated that global climate change could reduce global GDP by 20%. (Stern et al. 2006) Indeed, the threat is serious enough that claims for not growing the economy anymore or even “degrowing” are being strongly argued (Kallis 2011; Kallis et al. 2012). Some argue that reduction as high as a factor of ten is needed in affluent societies (Pretty 2013).

Moreover, it is generally believed that there is a fundamental tradeoff between societal energy consumption and self-interest at the individual and national level. It is believed that substantially reducing energy consumption requires individual sacrifices: It is widely (although not universally) believed that increased energy consumption is necessary both for overall national economic growth and to increase the directly consumed amenities (such as cooling, travel, lighting) that increase well-being (Jorgenson et al. 2014; Stern and Kander 2012; Dietz 2015; Carter 1977; Smil 2005). However recent research in the field of subjective well-being (SWB) offers theoretical reason for hope – the key point from SWB literature relevant here is that there is a consumption threshold beyond which SWB become satiated (Lamb and Steinberger 2017). In what follows we bridge the study of energy consumption and SWB and examine the empirical relationship between these at a global scale.

While energy consumption is useful, most energy consumption both pollutes and depletes natural resources (Arrow et al. 2004; Soytaş et al. 2007). How energy use affects human well-being on the whole remains an open question, which we examine in this research. Traditional approaches to examining the effects of development have measured societal benefits using the Gross Domestic Product (GDP) and its adjustments – per capita and purchasing power parity (Jorgenson 2014). Relying solely on

GDP as a measure of societal benefit has a broad range of shortcomings which are now becoming broadly recognized – and does not generally capture individual's evaluation of their quality of life (Giannetti et al. 2015). We evaluate the benefits and problems of energy use using both GDP and gold-standard measures of subjective well-being (SWB), life satisfaction.¹ Combining these measures reveals surprising relationships among energy consumption and societal benefit.

By combining data on energy consumption and SWB we find that the people in economically developed areas consuming more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation—it applies to patterns of energy consumption at the local, national, and global scale. Further, across both developed and developing countries with normal economies we observe a strong inverse relationship between SWB and the energy-intensiveness (the amount of energy per unit of GDP). This suggests that a national increase in energy consumption does not contribute to SWB (likely the opposite) except where such consumption leads to unusually great increases in GDP. By combining data on energy consumption and SWB we find that the people in economically developed areas that consume more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation—it applies to patterns of energy consumption at the local, national, and global scale.²

Analysis and Results

Selection of Measures and Data

There is a consensus that the survey items that we use are good measures of subjective well-being (Diener 2009; Oswald and Wu 2009; Stiglitz et al. 2009). All SWB measures come from surveys representative of given areas. Such measures are reasonably valid and reliable (Diener et al. 2013). One caveat is limited cross-cultural comparability (Diener and Suh 2003). Our cross-country results showed strong relationships and it is unlikely that the whole effect is due to measurement error, and we mostly used data within the US.

Unlike SWB, energy consumption measures used here are not very similar or directly comparable and differ across studies due to availability of measures. For the purposes of determining broad and durable patterns, we identified the existing data sources that have been systematically and continuously measured over periods of at least five years; have coverage of units within the target population (states, counties, nations) that is as complete as possible; and are of high quality.

¹ The concept of subjective well being (SWB) has been well-studied, for recent reviews see Diener (2009) and Diener et al. (2013). It is now widely recognized in this field of study that SWB is a broad multidimensional concept that unifies such popular concepts as (affective) happiness, and life satisfaction. Because it is a multidimensional concept, a complete measurement of all aspects of SWB requires collecting a large set of measures from each subject. Most broad-scale research in this field aims to measure the cognitive dimension of SWB, i.e., life satisfaction, because of its general domain (necessary for cross-comparisons), and established validity and reliability (see, Diener et al. 2013). We follow the standard practice in this area, and make use only empirical data and use life-satisfaction as our measure of SWB.

² As argued later, only developing nations could improve SWB through greater energy consumption, but across the developed world, the relationship between energy consumption and SWB is nil.

Global Patterns We started by examining the relationship between energy use and SWB across the world, by country. We used data from the World Database of Happiness (http://www1.eur.nl/fsw/happiness/hap_nat/nat_fp.php?mode=8). The SWB scale is based on multiple data sources and is for the most part based on responses to questions of the form: “All things considered, how satisfied are you with your life as a whole these days?” on a scale from 0 = “dissatisfied” to 10 = “satisfied.” The measure of total per capita energy consumption (kg of oil equivalent per capita) comes from the World Bank (<http://data.worldbank.org/indicator/EG.USE.PCAP.KG.OE>). All data were averaged over the years 2000–2009.

Data are plotted in Fig. 1. The basis for the received wisdom that increasing well-being requires increased energy consumption is illustrated by the figure on the left. Across countries there is a clear positive association between energy use and SWB. SWB generally increases alongside energy consumption, but the variance across countries is large and there are many outliers. For example, some countries that have high energy use, such as Russia (RU), are unhappy. In contrast, many countries such as Costa Rica (CR) or Mexico (MX) are able to reach the highest level of SWB while maintaining very low energy use. As energy use rises, its relationship with SWB flattens out,³ especially after the threshold of 5000 kg of oil equivalent per capita. In general, developed countries have a greater SWB (Mazur 2011; Jorgenson 2014).

How does the relationship between energy use and SWB change if we take economic development into account? It is well known that the relationship between energy use and SWB is affected by level of development (e.g., Jorgenson and Givens 2015; Knight and Rosa 2011). As recently highlighted by Dietz et al. (2009), it is important to examine how efficient a nation is in producing human well-being. Surprisingly, we find that when we measure energy as a function of economic efficiency, the relationship reverses. This is shown in the second panel of Fig. 1, which displays on the x axis the energy intensity of gross domestic product (energy/GDP). Countries that consume less energy per unit of wealth are happier. Some of the happiest countries are the highly-developed Nordic countries (DK, FI, NO, SE). Their energy consumption is high (Fig. 1, panel 1), but relative to income, their energy consumption is among the lowest in the world (Fig. 1, panel 2). In a descriptive sense, high energy intensity means that a country requires a high cost to convert energy into GDP. Another way to put this is that some countries are more efficient (in the economic, not technical sense) at converting energy into wealth.

There are a number of plausible causal explanations for this observed pattern. One hypothesis consistent with the data is that energy-use is largely unrelated to SWB once basic needs are met, but is driven up by other factors (some of which are explored below). A second plausible hypothesis is that countries with lower energy-intensity of GDP employ more energy-efficient technologies: These technologies reduce energy input for the same output of goods or services, so that consumption can remain high at low levels of pollution. A third plausible hypothesis is that countries with lower energy-intensity of GDP have economies that are rely less on inherently energy-intensive industries. A fourth plausible hypothesis is that country-specific characteristics (e.g. as industrial development, financial development, air temperature, and population) simultaneously drive energy use up and well-being down.

³ This is known as saturation, first mentioned with respect to human development by Martinez and Ebenhack (2008).

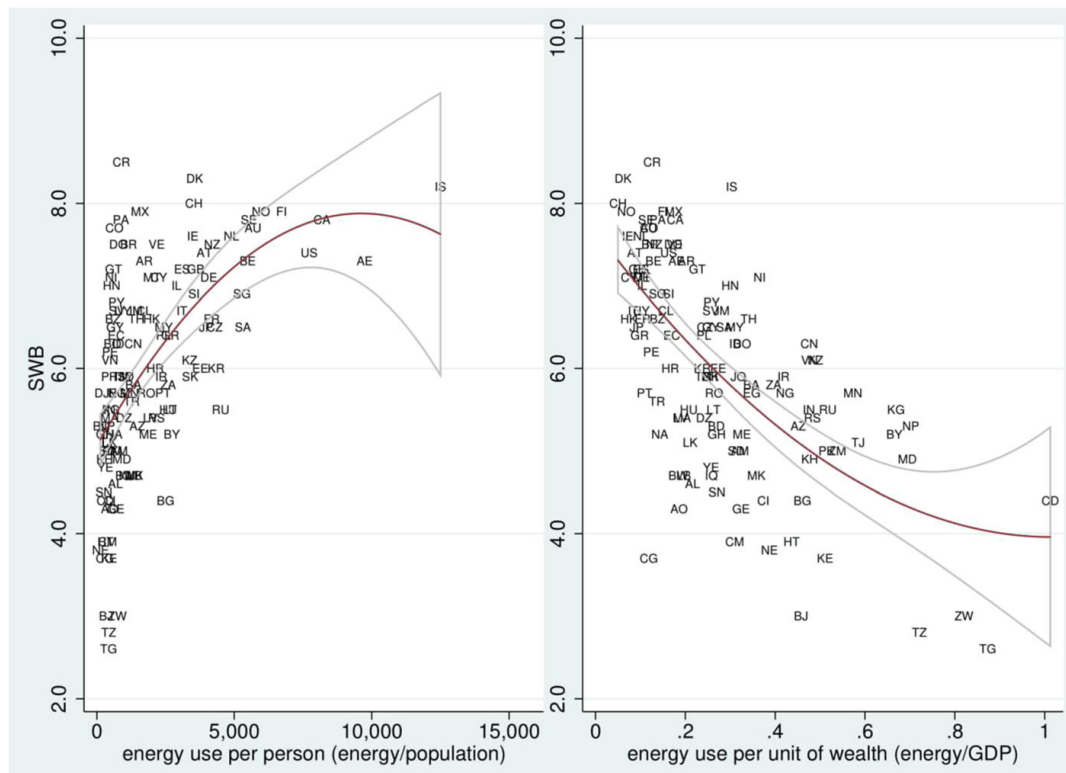


Fig. 1 Energy is measured as kg of oil equivalent. Countries that consume more energy per person are happier (first panel). The positive relationship in first panel is spurious—taking into account the level of development, the relationship reverses—the more energy per unit of wealth, the less SWB (second panel). Quadratic fit shown with 95% confidence intervals. Energy use refers to use of primary energy before transformation to other end-use fuels (kg of oil equivalent). All data were averaged over the 2000–2009 period. Country codes are in Table S1 in supplementary material. Several outliers on energy, GDP, or energy/GDP were dropped: Ukraine, Mozambique, Turkmenistan, Uzbekistan, Ethiopia, Luxembourg, and Qatar. And two additional outliers Kuwait, and Trinidad and Tobago—they would change L-shaped relationship in the second panel to U-shaped because they have extreme leverage—compare with Fig. S2 using full sample in supplementary material. Also, the relationships between income, energy and SWB are further explored in Figs. S3–S8—examining these interrelationships is a topic for the future research—present study focuses on documenting the main relationship only

These hypotheses cannot be definitely tested with the available data, however examining individual countries favors the first hypothesis: The Netherlands (NL) is rich, energy efficient, and happy. But there are also outliers. For instance, Colombia (CO) is happy and energy efficient, but poor. In general, Latin America poses a puzzle for SWB researchers. Latin Americans are relatively poor but happy. They also use very little energy and have similar energy intensity to that of the US. Notably, all Latin American countries cluster at the top left in the first panel. Great SWB is possible using little energy. East European post-Soviet countries, on the other hand, cluster at the bottom and some at the left. Some countries are relatively unhappy despite low energy intensity of GDP such as Congo (CG). Some countries, on the other hand, are relatively happy despite high energy intensity such as Trinidad and Tobago (TT) and Kazakhstan (KZ).

US States and Counties To further probe the hypotheses discussed above, and to answer old question of whether more energy is needed to increase well-being if there is already a great deal of energy being consumed (Mazur and Rosa 1974), we turned to the US, since it is among the countries that use the most energy per capita. State and

county level SWB data come from the Behavioral Risk Factor Surveillance System (<http://www.cdc.gov/brfss>) using a similar measure of SWB: “In general, how satisfied are you with your life?” on a scale from 1 = “very dissatisfied” to 4 = “very satisfied.” State energy data came from the US Energy Information Administration (<http://www.eia.gov/state>) and is measured as total energy consumption per capita in the residential sector.⁴ California’s residential electricity consumption per capita came from the Energy Consumption Data Management System (<http://www.ecdms.energy.ca.gov/elecbycounty.aspx>).⁵ State level data were averaged over 2005–2010; and California data were averaged over 2006–2010. Results are shown in Fig. 2.

Energy-hungry states are not happier. There are two outliers, Hawaii and California, consuming much less energy in the residential sector than others. We zoom in on California counties in the second panel of the same figure. There is also a great deal of variation in energy use across California counties, and the relationship with SWB is also nil. We have experimented with energy intensity of GDP as we did earlier across countries, but in the case of the US subregions the results are not different.

Over Time Movement It is well-known that SWB is related to income in a cross-section, but not over time (Easterlin 1974; Easterlin et al. 2012). To account for this we supplemented our cross-sectional results with an exploration of SWB and energy consumption over time. We used the General Social Survey (<http://gss.norc.org>), which measures SWB as follows: “Taken all together, how would you say things are these days—would you say that you are very happy, pretty happy, or not too happy?” 1= “not to happy”, 2=“pretty happy”, 3=“very happy”. Energy consumption is measured as total energy use per capita in the residential sector, the same measure as used for states. Figure 3 shows SWB and energy use over time by census division. There is not much co-movement: the two series correlate at .2 only. Energy consumption is only weakly related to SWB.

High residential energy use in the US is often linked to energy inefficient suburbanization: sprawling car-dependent large housing (so called McMansions) (Duany et al. 2001). Energy use and pollution per capita are higher in sprawling suburbs than in compact cities (Meyer 2013). Such suburbs typically crop up in affordable and spacious South and indeed much of the country, and least in increasingly compact and expensive North East and West Coast⁶—Figure 3 agrees: New England and Pacific have their energy use per capita declining. Another problem for climate change is that the rest of the World may want to mimic this unsustainable lifestyle.⁷

Americans continue to consume large amounts of energy as compared to other countries. With the notable exceptions of New England and the Pacific region, energy

⁴ The series code is TERPB, and further information is available at https://www.eia.gov/state/seds/sep_use/notes/use_technotes.pdf

⁵ Total residential electricity consumption has two components: (A) validated Quarterly Fuel and Energy Report (QFER) retail sales/delivery data and (B) self-generation. Electricity Consumption = retail sales + self-generation. Retail sales: electricity amounts delivered from the grid and consumed by the customer. Self-generation: electricity produced and consumed on-site by the customer (for example, parking lot roofs covered with PV panels). Self-generation only contributes to total consumption and not to total sales.

⁶ Parts of California, notably Los Angeles, are often argued as examples of wasteful suburbanization, yet Los Angeles new homes are much smaller than average (<https://www.laweekly.com/news/how-the-size-of-the-typical-la-home-has-grown-over-the-years-7364336>).

⁷ Notably, Central and Eastern Europe have been suburbanizing recently (Stanilov and Sýkora 2014).

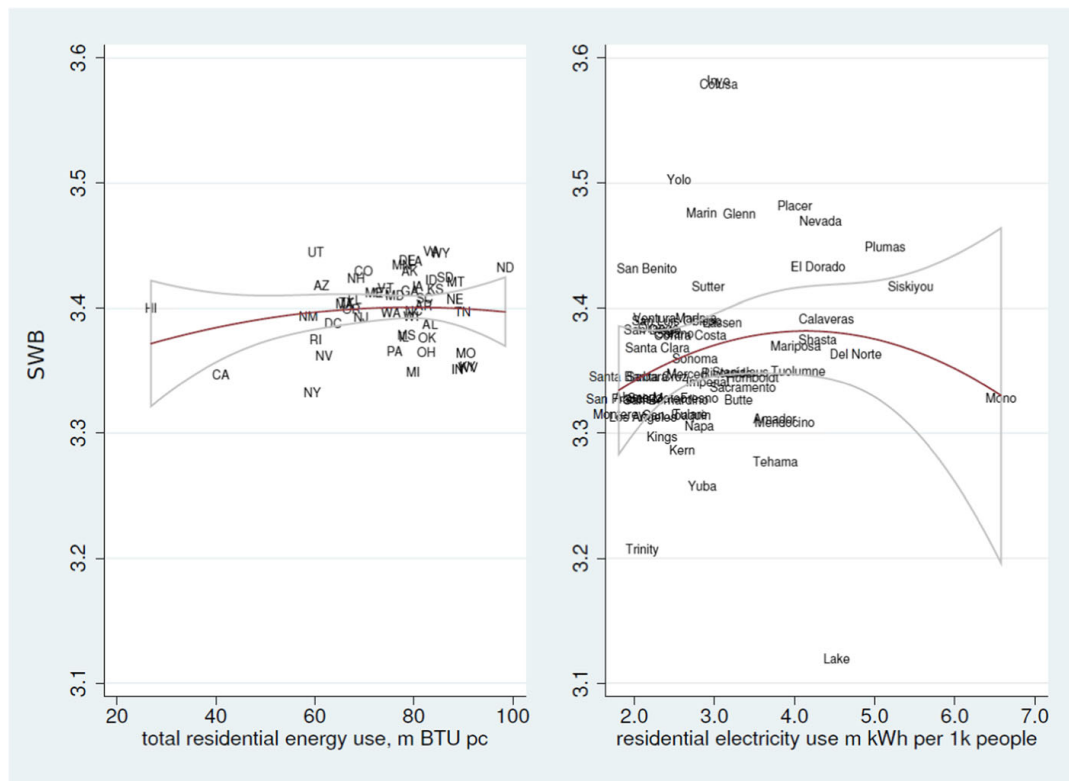


Fig. 2 SWB and total residential energy use across US states and residential electricity use across California counties. The relationship was also quite flat in terms of total energy consumption and its GDP intensity. State level data were averaged over 2005–2010, and California data were averaged over 2006–2010

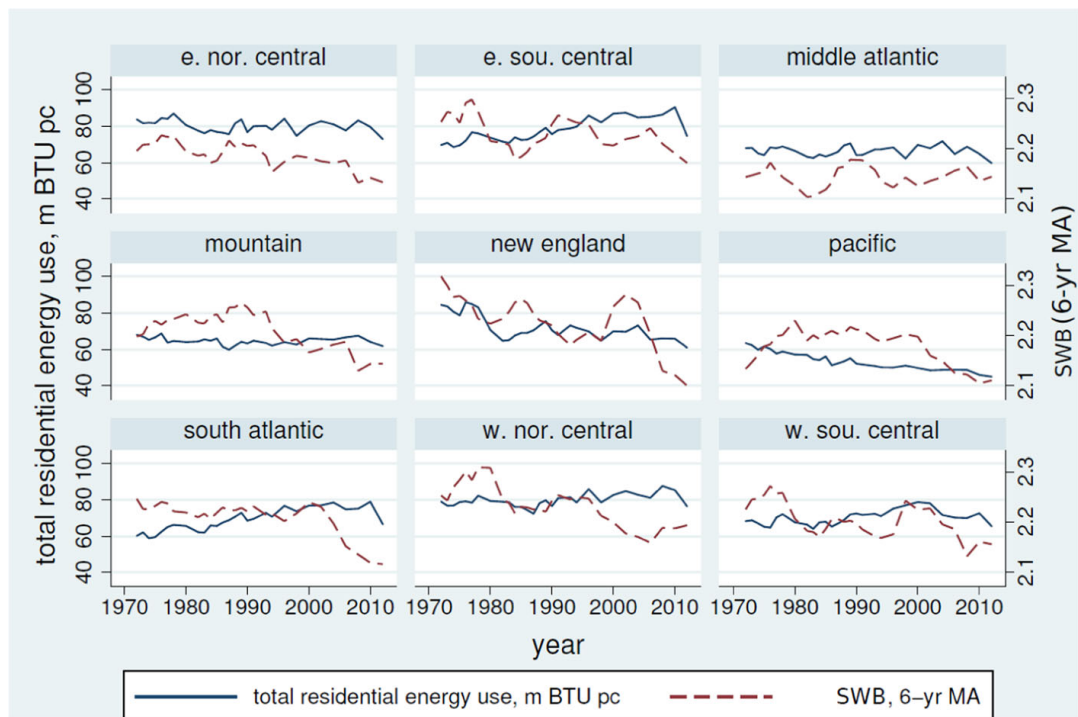


Fig. 3 SWB (6-yr moving average) and total energy consumption in residential sector per capita. Correlation is .2 only (for unsmoothed series)

use is not decreasing, and sometimes it is increasing. Americans do not spend any less on energy either: 5–10% of personal expenditure over past 50 years were on energy (BEA 2014).

Discussion

The analysis above shows that energy use is apparently unrelated to SWB, especially in developed nations such as the US. Why? There are a number of potential causal explanations rooted in on existing theories of individual behavior, social behavior, and economics. Existing observed information does not allow us to definitively test each of these causal theories here. The discussion below aims to suggest productive areas for future by outlining four possible behavioral explanations, and several additional economic/industrial explanations.

One possibility is that bounded rationality plays a strong role: Decisions on how much to produce and consume based on expected SWB are tricky, humans are often predictably wrong and experience much less SWB than expected (Kahneman et al. 1997), and are wrong in their perceptions of energy consumption and savings (Attari et al. 2010; Dietz 2014). Thus it is perhaps not surprising that there is a discrepancy between expectation (more energy use, more SWB) and experience (more energy use, no more SWB) is not surprising.⁸

Second explanation is that energy consumption is a positional good: Consumption buys position, but because everyone is in competition with everyone else, the race cannot be won (Frank 2012; Kasser 2003). A related explanation is a “hedonic treadmill” (Brickman et al. 1978): It is possible that consumption does not increase well-being because of hedonic adaptation. More consumption would not make people happy if their basic needs are already satisfied.

A third potential explanation builds on Veblen’s concept of conspicuous or wasteful consumption (Veblen 2005a, b). Such consumption does not satisfy needs but simply aims to demonstrate that one is better than others. Much of such consumption wastes energy without satisfying human needs, and does not make us any happier (Csikszentmihalyi 1999; Frank 2004, 2005, 2012).

A fourth potential explanation based on the livability theory (Veenhoven 2014), is that consumption result in more well-being only up to the point that is used to satisfy basic human needs. Hence, energy used to alleviate extreme temperatures, provide food, shelter, basic transportation, etc, will increase human SWB (if it out-weighs the costs such as pollution). But energy consumed on non-essential human needs (arguably most energy consumed in developed nations) will result in little SWB, if any, and our findings are consistent with such an explanation.

In parallel with these potential behavioral explanations, industrial and economic factors could contribute to the relationship between energy use and SWB. One possibility that any direct positive effect of energy use on SWB is offset by an increase in pollution. (See, for example, Li et al. 2018 for a review of the effects of air pollution on

⁸ While we believe the behavioral explanations are pivotal, there are other potential explanations. For instance, access to technology may affect the threshold energy level needed to meet core human needs.

SWB). If such an indirect effect exists, the relative dependence of a nation's economy on industries that are energy intensive and/or highly polluting, and the availability and deployment of energy efficient technologies in these industries, would be likely to mediate together the relationship between SWB and energy consumption (Stern 2011). These mediators in turn may be causally affected by country-specific characteristics such as economic history, population, and climate. Future research can focus on testing specific explanations—here we have focused on documenting the overall relationship.

Conclusions

Again, while for simplicity we use term “energy”, it is important to keep in mind that we have used different measures of energy. At the country level, among developed countries, there is no relationship between energy use and development. Further, for all countries with normal economies, the less energy-intensive is a country's economy, the happier are its people. (Fig. 1, Panel 2). Across US states and California counties, energy consumption and SWB have no apparent relationship. Likewise, the changes over time in energy use are almost unrelated to changes in SWB across US Census regions. With this correlational study we aim to bring the relationship between energy use and well-being to wider audiences, and encourage more research in this area.

While correlation is not sufficient for causality, it is typically necessary – especially across relatively homogeneous units of observation (Mazur 2011). Although this is an observational study and does not demonstrate causality, the patterns observed suggest that in the US, and across the developed World, change in energy use will not cause significant change in SWB. Further it suggests that, even in developing countries, increases in energy consumption do not contribute to SWB except where such consumption leads to unusually great increases in GDP. These patterns contrasts with widely held assumption about the relationship between SWB and energy consumption. There are many possible explanations—and while the available data does not support a causal analysis, we offer some conjectures for future research.

We found that energy consumption is neither necessary for SWB, nor linked directly to it, and we speculate the relation between energy use and SWB is very similar to the relation between economic growth and SWB, i.e., the Easterlin Paradox (Easterlin et al. 2010): common wisdom has it that over time income should increase SWB, and so common wisdom has it that energy use should increase SWB, but neither does. Specifically, Easterlin Paradox states that at a point in time SWB varies directly with income, but over the long run (10 years or more) SWB does not increase when a country's income increases. Two countries that dramatically increased both GDP and energy use are Japan and more recently China. From 1960 till 1990 Japanese energy use per capita increased more than three fold; from 1990 to 2007 Chinese energy use per capita more than doubled, but SWB remained flat in both China and Japan.⁹

⁹ Primary energy use (before transformation to other end-use fuels) in kilograms of oil equivalent, per capita from World Development Indicators accessed through <https://www.google.com/publicdata>. SWB data come from Frey and Stutzer (2002) and Easterlin et al. (2012)—note that Chinese SWB slightly increased in last several years. Also note that in the US both energy use per capita and SWB remain flat over past four decades (using GSS data and above measure of energy use). We provide additional analyses in supplementary material.

It is striking that there were only several attempts to relate energy consumption to SWB. In a small sample of 55 countries, one early study used a set of 27 indicators to measure quality of life (Mazur and Rosa 1974), which was later extended over time (Mazur 2011). Mazur (2013) confirmed his earlier findings: among industrial nations, already high in their energy and electricity consumption, further increases in per capita energy or electricity consumption are unrelated to changes in SWB. Another study is a recent article that analyzes cross national data (Winfrey 2013). But neither study explores energy intensity of GDP, nor variation at finer geographic representation than a country. This is an important contribution of this study: there is more heterogeneity across units at more aggregated levels (e.g., countries are less comparable than US states), and there is less precision (e.g., state level SWB and energy averages are less precise descriptions of individuals than county level averages). And it is well known that there may be different relationships at different levels of aggregation (e.g., Ashkanasy 2011), i.e. earlier cross-national findings may not hold at state or county levels. Furthermore, cross-sectional relationships often differ from time-series relationships (Easterlin et al. 2012), i.e. earlier cross-sectional findings needed confirmation in time-series analysis in present study.

We do not claim causality in this correlational study. Yet, as persuasively pointed out by labor economist Andrew Oswald (e.g., Blanchflower and Oswald 2011; Oswald 2014), correlational studies are not without merit despite what many economists think—many scientific breakthroughs were first discovered in observational studies, for instance, that smoking is related to cancer. It is often overlooked that experiments tend to suffer from many problems that are not inherent in observational studies such as lack of external validity, small sample size, artificial laboratory setting, and so forth, for discussion see Pawson and Tilley (1997). At the same time, we encourage research into the causal relationships between energy use and SWB.

There is a need for future research in this important area. There have been many calls to systematically collect SWB data, and we should collect energy use data for the same subjects. Such data would allow the exploration of the relationship at the individual level.

We conjecture that in the developed world, we can decrease energy consumption without much loss in SWB, if any. While people from India and China may need to consume more energy, it is possible that Americans could consume considerably less without reducing SWB. For example, Texans might reduce energy consumption by half, following the example of California, and remain content. We agree with Dietz et al. (2009) that we should strive for environmentally efficient well-being: achievement of human well-being at low level of environmental stress; and we agree with Steinberger and Roberts (2010) that high human development can be achieved at moderate energy and carbon levels.

Based on the our analysis, we suggest an interventions to decrease energy consumption. First, we may simply need to increase awareness of the empirical relationship between SWB and energy consumption: there is no evidence that increasing already substantial consumption does buy substantial SWB. Just as increasing income beyond a point does not result in much SWB (Kahneman and Deaton 2010), increasing energy consumption beyond a point does not result in much SWB either. In short, SWB can be achieved at low levels of energy consumption—human flourishing requires energy to satisfy basic needs only. We are hopeful that awareness and education can change behavior. Many other ways to curb consumption have been suggested (Dietz 2014, 2015; Asensio and Delmas 2015; Dumas 1987; Attari et al. 2010). Proposals to

consume less energy are urgently needed as market economy is often incentivizing people to consume more energy, for instance when new sources are discovered as with current natural gas bonanza in the US (Hopkins 2017).

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Compliance with Ethical Standards This material has not been published in whole or in part elsewhere; the manuscript is not currently being considered for publication in another journal; all authors have been personally and actively involved in substantive work leading to the manuscript, and will hold themselves jointly and individually responsible for its content.

Conflict of Interest No conflict of interest declared.

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4 papers: urban-rural happiness gradient

4.1 Urbanism and happiness: A test of Wirths theory of urban life (Okulicz-Kozaryn and Mazelis 2018)

Urbanism and happiness: A test of Wirth's theory of urban life

Adam Okulicz-Kozaryn

Rutgers-Camden, USA

Joan Maya Mazelis

Rutgers-Camden, USA

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Abstract

Social scientists have long studied the effects of cities on human wellbeing and happiness. This article demonstrates that people in cities are less happy, confirming a long-standing argument in the literature. But it had not yet been tested whether it is urbanism that negatively affects happiness, or if urban problems such as crime and poverty are to blame. Wirth posited that urbanism itself led to negative effects, but Fischer noted the necessity of empirical tests of Wirth's ideas. This study uses a happiness measure to provide a new look at the old question of urban unhappiness. Using the Behavioral Risk Factor Surveillance System from the US Centers for Disease Control and Prevention, we aim to untangle the effects of the city itself and urban problems on happiness in the USA. We find that the core characteristics of urban life (in particular size and density) contribute to urban unhappiness, controlling for urban problems. Urban unhappiness persists regardless of urban characteristics.

Keywords

city, happiness, life satisfaction, subjective wellbeing, urbanism

摘要

社会科学研究者一直在研究城市对人类福祉和幸福感的影响。本文证明城市里的人们确实不那么幸福，印证了研究文献中长期存在的一个论点。但到底是都市生活对幸福感产生了负面影响，还是应归咎于犯罪和贫困等贫困问题，这一点尚未得到验证。沃斯 (Wirth) 认为都市生活本身导致了负面影响，但费歇尔 (Fischer) 指出有必要对沃斯思想进行实证检验。本研究采用了一项幸福度指标来重新审视都市生活之不幸感的旧问题。我们使用美国疾病控制和预防中心的行为风险因素监测系统，旨在厘清城市本身和城市问题对美国幸福感的影响。我们发现，在假定城市问题不产生影响的情况下，城市生活的核心特征（特别是规模和密度）促成了城市中的不幸福感。不论城市特征如何，城市中的不幸福感都会持续下去。

关键词

城市、幸福、生活满意度、主观幸福感、都市生活

Corresponding author:

Adam Okulicz-Kozaryn, Rutgers-Camden, 401 Cooper St,
Camden, NJ 08102, USA.

Email: adam.okulicz.kozaryn@gmail.com

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This article addresses a fundamental social science question: how does the modern world affect those who live in it? Modernisation and industrialisation brought urbanisation, rife with complex problems. Tönnies ([1887] 2002) and Durkheim ([1893] 1997) challenged us to investigate what living in cities does to us. The first American sociologists, Louis Wirth among them, were urban sociologists, similarly concerned with city characteristics, urban problems and happiness (Wirth, 1938). They did not use the term happiness, but other terms to denote the negative side of the urban way of life, such as ‘malaise’.¹ Wirth’s theory of urban life focused on how urbanism led to various negative consequences, including (1) cognitively, in terms of alienation, (2) behaviourally, in terms of deviance, and (3) structurally, in terms of anomie (normlessness), all of which would lead to unhappiness.

There is a long-standing debate about whether people in cities are unhappy. The academic debate started in sociology (Durkheim, [1893] 1997; Park, 1915; Weber et al., [1921] 1962; Wirth, 1938), but was abandoned by sociologists in the 1970s after Fischer’s major works were published. Recently, the topic was taken up by other disciplines, yet it has not addressed sociological research, and has used other approaches and foci. Regardless of the fact that urban life has much to offer residents, there is a consensus emerging among scholars that city living makes us, on average, unhappy and unhealthy (Adams, 1992; Adams and Serpe, 2000; Amato and Zuo, 1992; Balducci and Checchi, 2009; Berry and Okulicz-Kozaryn, 2009, 2011; Evans, 2009; Lederbogen et al., 2011). Those authors, however, used small data sets with few variables, typically operationalised city simply as the size of settlement, or used a dichotomous urban/rural variable. No study has controlled for city-level variables such as per capita income; no study so far has used a

model that accounts for both personal and city-level factors that affect happiness. A fundamental question as framed by Fischer (1972, 1973, 1975) remains unanswered: is it cities themselves, in terms of size, density, and heterogeneity, that lead to unhappiness, or is it problems people typically associate with cities – notably poverty, crime and lack of support – that make people unhappy?

The first urban sociologists proposed that urbanisation created malaise (unhappiness) because of the core characteristics of cities: increased population size created anonymity and impersonality, density created sensory overload and withdrawal from social life, and heterogeneity led to anomie and deviance (see Park et al., [1925] 1984; Simmel, 1903; Tönnies [1887] 2002; Wirth, 1938). Some might assert that cities are defined by their problems, and argue that measures can be used interchangeably to examine both urbanism and urban problems. However, we follow Park, Wirth, and others to define urbanism by using population size and density. We seek to advance the understanding of urban unhappiness by better measuring urbanism using several size and density variables, and controlling for urban characteristics and problems to test whether urban unhappiness persists regardless of them. Recent large-scale survey data representative of a wide range of cities, as well as counties that are not cities as a comparison group, make this test possible.

Theoretical background

There are many advantages to cities. Labour specialisation, economies of scale, invention and creativity are all made possible by high-density living (e.g. Florida, 2008; Glaeser, 2011; O’Sullivan, 2009). Human civilisation as we know it was made possible by cities and the labour specialisation that took place in them (O’Sullivan, 2009); this made organic solidarity possible (Durkheim,

[1893] 1997). Average standard of living has improved tremendously, even with high levels of inequality: by one estimate, those in the bottom decile today have a better standard of living than all but the top 1% did 100 years ago (Bok, 2010). For an overview of the progress that Americans have made since colonial times, see Fischer (2010). Progress in standard of living and urbanisation are closely related, leading some scholars to conclude that we are happier in cities than we are elsewhere (Glaeser, 2011); this is not so. In fact, Americans have not become any happier over time (Easterlin, 1974), and we are, on average, the least happy in cities (Berry and Okulicz-Kozaryn, 2011). The most celebrated thinkers in America's intellectual history almost universally expressed ambivalence or animosity toward the city, among them: Jefferson, Emerson and Thoreau (White and White, 1977). Wirth (1938) famously summarised the negatives of city life, building on the worries of earlier European sociological theories (Tönnies, [1887] 2002). Cities are collections of heterogeneous people with different specialisations; there is a lot of variability among city dwellers. Relations between them are anonymous, superficial, impersonal, transitory, unstable and insecure (Wirth, 1938). Early theorists lamented what they saw as a lack of norms, which led to moral deviance such as crime (Wirth, 1938). Cities are fast-paced and crowded, creating a psychological overload because of the intensification of stimuli for urban dwellers and occasioning detachment, and a retreat to social isolation (Simmel, 1903). Simmel argued that people necessarily adapted to urban life by developing a blasé attitude: with the rapid pace and all the people we come across, we would not be able to handle it if we had to deeply engage with everyone and everything, so we become detached. Simmel noted that this reserve provides more personal freedom and privacy to urban dwellers, but it also necessitates

impersonal interactions with those we encounter (Simmel, 1903). Cities generate moral *disorder*: misunderstandings, tension and conflict (Smelser and Alexander, 1999). For more elaboration see Fischer and Merton (1976).

Recent evidence raises other concerns about cities: density and heterogeneity predict lower social capital (Helliwell and Wang, 2010) and low social capital is associated with poor health and higher mortality (Kawachi et al., 1997; Subramanian et al., 2002). Poverty is concentrated in cities, and the more concentrated the poverty, the worse it is for people (Jargowsky, 1997). There is an abundance of crime (Bettencourt et al., 2007, 2010) and segregation (Jargowsky, 1997; Massey and Denton, 1993). There is more crime in cities than there is elsewhere, and high crime predicts poor health (Lynch et al., 2004; Zimmerman and Bell, 2006).

High density predicts low happiness (Fassio et al., 2013; Lawless and Lucas, 2011). One specific reason why there may be less happiness in cities is that 'the pecuniary nexus tends to displace personal relations' (Wirth, 1938: 1). In addition, 'fashion tends to take the place of custom' (Park, 1915: 605), and consumption does not have a lasting impact on life satisfaction because people adapt to material goods (Easterlin, 2003). Recent neurological evidence confirms early sociological theory: great cities actually overstimulate our brains to the point where it is not healthy (Lederbogen et al., 2011). Schwartz (2004) shows that the abundance of choice, exemplified in cities, often leads to depression, loneliness, anxiety and stress. Americans do not merely prefer smaller areas as documented earlier by Fuguitt and Zuiches (1975) and Fuguitt and Brown (1990), they are also happier in smaller areas (Lawless and Lucas, 2011).

There have been many attempts at discerning the effect of city living on happiness. But the limitation of even the most recent

research is that it does not control for city-level variables such as per capita income, and city disadvantages such as crime and poverty, something that Fischer suggested researchers should do over 40 years ago. Fischer (1973: 222) acknowledged that people are less happy in the biggest cities, but he was not sure whether it was urbanism or other confounding variables that make residents unhappy in cities:

(1) Such preferences [for rural living] may be a function of idealized images founded on popular conceptions of urban and rural life. (2) They may indicate utopian hopes of maintaining urban opportunities in small communities. [...] (3) These evaluations may result from the contemporary state of American cities rather than from the nature of cities per se.

The remainder of this article will try to answer item (3) above. We aim to discover whether city residents are less happy, controlling for problems people typically associate with contemporary cities. If so, these problems cannot be the cause of lower happiness in cities. Rather, urbanism's core characteristics – large numbers of people and high density – along with other factors, make city dwellers unhappy:

H1: people in cities are less happy than are those in other areas if cities are defined by:

H1a: population size

H1b: density

H2: urban unhappiness will persist after controlling for problems people typically associate with cities.

In testing the above hypotheses we control for city characteristics that may affect happiness and for known person-level predictors of happiness.

Data and measurement

We analyse the 2005 Behavioral Risk Factor Surveillance System (BRFSS) from the

United States Centers for Disease Control and Prevention. The BRFSS is representative of the USA, with a total sample size of over 100,000 people per year, well suited for the study of cities. The regular BRFSS data set, however, is not representative of most counties, so we use the SMART (Selected Metropolitan/Micropolitan Area Risk Trends) version of the BRFSS that is representative of counties; for simplicity we will refer to it as the BRFSS. The downside of the SMART version is a smaller number of counties, 232, but it still provides a good sample of the USA with enough variation.² See the appendix (available online) for a listing of counties. In addition to the 2005 BRFSS, as a robustness check, we include models using data for 2006, 2007 and 2008 in the appendix (available online), because the more recent waves have bigger samples, and to account for potential lag effects from variables measuring urban problems at the county level. In the main text we report the 2005 BRFSS results, because the county-level data we use are either for 2003, 2004 or 2005. All county-level data come from the Inter-university Consortium for Political and Social Research: County Characteristics, 2000–2007.

We use data at person and county levels. We do not have data at the city level. City attributes are measured at the county level; many metropolitan areas cover the whole county. Even where a metropolitan area covers part of a county, it seldom ends abruptly and changes into a non-urban area or open land. Where a metropolitan area abruptly ends, then it is due to a natural boundary such as an ocean in the case of coastal cities, but in that case the county ends as well. Furthermore, this limitation is ameliorated by another operationalisation of cities: the urban–rural continuum, which differentiates between metropolitan areas, cities, towns of various sizes and smaller areas. Another triangulation of measurement is performed

using a person-level location variable: metropolitan status code.

The person-level dependent variable: Happiness

As noted above, we treat malaise as unhappiness, following research by Fischer (1972, 1973), who pioneered using happiness data to study urban malaise. Yet data and statistical software did not enable a proper test in the early 1970s. We use large-scale data and a model controlling for both personal and ecological characteristics.

Happiness is a multidimensional concept (e.g. Hills and Argyle, 2001), though for simplicity we use the terms happiness, life satisfaction and subjective wellbeing interchangeably (the dimensions overlap). The BRFSS has only one survey item measuring happiness.³

The survey item reads ‘In general, how satisfied are you with your life?’. For simplicity, we recoded answers so that a higher numeric value means more happiness: 1 is ‘very dissatisfied’, 2 ‘dissatisfied’, 3 ‘satisfied’ and 4 ‘very satisfied’. Over 90% of respondents were either satisfied or very satisfied with their lives. The same variable has been used in recent happiness studies by Oswald and Wu (2010, 2011), Glaeser et al. (2014) and Winters and Li (2015).

The happiness measure, even though self-reported and subjective, is reliable, valid (Myers, 2000), and closely correlates with similar, more objective measures such as brain waves (Layard, 2005). Unhappiness strongly correlates with suicide incidence and mental health problems (Bray and Gunnell, 2006). Finally, to be clear, we study here general/overall happiness, not a domain-specific happiness such as neighbourhood or community satisfaction.

The main person- and county-level independent variables: City size/city population

In this study, we sacrifice breadth of measurement of urbanism at the cost of precision and use the most precise measures available: population size and density. Population size is used in two ways: as an urban–rural continuum with nine categories defined by the USDA (US Department of Agriculture) and as a metropolitan status code based on a survey item from the BRFSS. We chose continuous/ordinal measures because they convey more information than do binary or nominal ones (see Meyer, 2013; Wirth, 1938). The county-level urban–rural continuum ranges from 1 (‘metro>1 million’) to 9 (‘non-metro<2500, not adjacent to metro’).⁴

The metropolitan status code differs within a county; it is a person-level variable and reflects the urban dweller status more precisely than does the county-level urban–rural continuum. The metropolitan status code is broken down into the following dummy variables: ‘not in an MSA’, ‘inside a suburban county of the MSA’ and ‘outside centre OR in MSA without centre’. ‘Centre city of MSA’ is the base case, and MSA stands for Metropolitan Statistical Area, a geographical area associated with an urban area of 50,000 people or more, as defined by the United States Office of Management and Budget.

The main county-level independent variable: Population density

We measure density as population (in thousands) per square mile, and there are staggering differences; density ranges from four people per square mile in Sweetwater County, WY to 70,000 people per square mile in New York County, NY.

The measures of urbanism we use do not capture heterogeneity directly. We do not include heterogeneity for three reasons. First, it is not obvious which type of heterogeneity is to be measured. In modern times we think of racial, ethnic and class diversity, but in Wirth's conception, influenced by Durkheim's focus on the increasing division of labour and occupational specialisation that accompanied urbanisation, heterogeneity may have been more about occupational diversity; Wirth writes of heterogeneity in 'social groups' (1938). It also may have been more about diversity in nation of origin, which while related to contemporary understandings of ethnicity, is not synonymous with the latter. As Wilson (2012) and Finney and Simpson (2005) note, sometimes the focus on race and ethnicity is misplaced. So we might consider measures of ethnicity or national origin, race, religion, income or occupation, for example. Second, there is precedent in recent literature to omit heterogeneity as part of a definition of urbanism, though literature consistently considers size and density (e.g. Meyer, 2013). Third, we looked at various measures of racial diversity, but did not find a consistent and robust relationship.

County-level controls: Measures of urban characteristics

As we have argued above, this paper's key contribution is to disentangle the effects of the city (size, density) and urban problems on the happiness of city residents. To achieve that, we control for the following city characteristics: crime, housing stress, low education, low employment, persistent poverty, population loss, personal income (USD 1000)/cap, and percent Black. This set of controls addresses a fundamental confounding factor and an alternative explanation for city unhappiness: there are many poor people either stuck because they cannot afford to move, or who come to

cities seeking a better life. Poverty and crime are seen as fundamental urban problems (Jargowsky, 1997).⁵ Classical theorists posited that lack of support from others is an issue in cities (Park, 1915; Wirth, 1938), though Fischer (1982, 1995) argues that this urban problem is a myth.⁶ However, we control for lack of support with a person-level variable: social/emotional support.

We control for eight city characteristics, but there are countless characteristics one could hypothetically include. Therefore we use county fixed effects (we can measure city size at person level, because there is a person-level survey item asking respondents about the location of their residence). Finally, we focus on several of the happiest and least happy counties. We focus on issues thought to be major urban problems, such as poverty and crime, as defined in the literature (e.g. Jargowsky, 1997), as well as other variables that might have an effect, such as personal income and percent Black. 'Problems' such as congestion or high cost of living are arguably very closely related to population density, and hence are characteristics of urbanism as defined here.

Person-level controls: Predictors of happiness

We control for several variables because they predict happiness as shown in the literature. Exploring the interactions of these variables with city size and density may be an interesting topic for future research, but it is beyond the scope of this study.

Increased income boosts happiness and unemployment depresses it (e.g. Di Tella and MacCulloch, 2006; Di Tella et al., 2001a, 2001b). Married people are happier than those who are unmarried (e.g. Diener and Seligman, 2004; Myers, 2000). African Americans are less happy than are Whites in the USA (e.g. Berry and Okulicz-Kozaryn, 2009, 2011), and they are concentrated in

cities, so it is important to control for race in the context of urban unhappiness. Happiness is U-shaped with age, that is, both the young and the old are happier than are people during midlife (e.g. Sanfey and Teksoz, 2005).

Health is consistently an important predictor of happiness, so we control for health status, and we also control for education, which some authors find to predict happiness (see Dolan et al., 2008). There are substantial urban–rural differences in education. In general, urban relationships are impersonal and transitory (e.g. Wirth, 1938) and some posit that there is less support in cities (White and White, 1977) and less trust in cities (Berry and Okulicz-Kozaryn, 2011). We measure lack of support using the following question: ‘How often do you get the social and emotional support you need?’. It ranges from 1 (never) to 5 (always).⁷ Together with several county-level variables this accounts for urban problems in testing whether it is such urban problems – or cities themselves – that create urban unhappiness.

The appendix (available online) lists all the variables used in this study along with their definitions, and includes summary statistics as well.

Model and results

Our modelling approach is similar to Fernandez and Kulik (1981) who also tested the effect of place on happiness and found that people are happier in smaller areas. By comparison with their work, the sample in the present study is over ten times larger, uses additional robustness checks and controls for what are typically characterised as urban problems.

We use a standard OLS regression with clustered standard errors by county. We also use BRFSS sampling weights to account for oversampling. We treat the four-step happiness variable as continuous (see Blanchflower and Oswald, 2011; Ferrer-i-Carbonell and Frijters, 2004).⁸

Results are shown in Table 1. We first estimate a base model (A) that includes known predictors of happiness. Column (B) adds two basic proxies for urban problems: crime and poverty. We also include percent Black, not as an urban problem but because many US central cities include a large African American population, African Americans score lower on happiness measures than do white Americans, and therefore the lack of this variable in a model could bias results because of different levels of happiness between racial groups. Finally, column (C) includes several additional controls for city characteristics and possible urban problems: low education, housing stress, low employment, population loss and personal income per capita.

We start with a person-level location variable. City size dummies are listed first in the table and indicate difference relative to the centre city. The most significant, both statistically and practically, is the ‘not in MSA’ dummy. As hypothesised, people are happiest outside of cities. Estimates are stable across specifications; it does not matter whether the city has stereotypical urban problems. People living outside of metropolitan areas as compared with central cities are happier by 0.05 on a scale from 1 to 4. This difference may appear to be small in practical terms, but it is still important. First, it contradicts claims that people are happier in cities, notably those by Glaeser (2011). Second, happiness is largely due to person-level influences such as unemployment, and ecology is secondary. Across 232 counties in our sample, happiness ranges between 3.18 and 3.56, with the vast majority of counties, 209 of the 232, in the 3.30 and 3.40 ranges. Third, an ecological difference of 0.05 (coefficient on ‘not in an MSA’ from Table 1) on a happiness scale from 1 to 4 can translate to large effects: for example, all else being equal, if one-third of 1% of the US population, about 1 million people, who live in

Table 1. Survey weighted regressions of happiness.

	A	B	C
Outside centre OR in MSA without centre	−0.0001	0.0002	−0.0017
Inside suburban county of MSA	−0.0024	0.0098	0.0079
Not in an MSA	0.0467***	0.0445**	0.0451**
<i>County-level controls</i>			
Crime rate index		7.90e−06**	7.90e−06*
Persistent poverty		0.0061	0.0175
% Black		−0.0012**	−0.0009*
Low education			0.017
Housing stress			−0.0019
Low employment			−0.028
Population loss			−0.02
Pers. inc. (USD 1000)/cap			0.0001
<i>Person-level controls</i>			
Income	0.0255***	0.0260***	0.0259***
Married or member of an unmarried couple	0.1406***	0.1410***	0.1409***
Unemployed	−0.1687***	−0.1676***	−0.1672***
Age	−0.0078***	−0.0081***	−0.0081***
Age squared	0.0001***	0.0001***	0.0001***
White	−0.0558***	−0.0581***	−0.0558***
Education level	−0.0085+	−0.0110*	−0.0110*
Soc/emo support	0.1847***	0.1842***	0.1844***
General health	0.1342***	0.1357***	0.1356***
N	138,453	132,677	131,657

Note: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

central cities instead lived in non-urban areas, it would mean about 50,000 (0.05*1 million) people would be satisfied with their lives rather than dissatisfied.

All person-level variables are significant, while most county-level variables are not significant. It could be that the location variable is significant because of the large sample size at the person level. Yet, in Table 2 we use a county-level size variable, the urban–rural continuum. It ranges from 1 (‘metro > 1 million’) to 9 (‘non-metro < 2500, not adjacent to metro’). This triangulation of measurement serves as a robustness check; the smaller the place, the more happiness, no matter how the city is measured.

The coefficient estimates are 0.01, and there are eight categories on the urban–rural continuum, so a change from 1 (‘metro > 1 million’) to 9 (‘non-metro < 2500, not

adjacent to metro’) is associated with a happiness difference of about 0.1. What is striking about this second set of results is the persistence of statistical significance at the $p < 0.01$ level on the urban–rural continuum variable; the significance of all county-level control variables is much less stable. Urban unhappiness exists, and hence we find support for hypotheses H1a and H1b. To account for possible lag effects from contextual, county-level variables on happiness, we reran these models using the 2006, 2007 and 2008 BRFSS data (the results are in the appendix, available online).

Finally, Table 3 shows that the denser the county, the less happy people are there. Adding controls attenuates the effect. Crime and poverty explain away some of the city effect. Percent Black also explains some of the city effect. Yet, density remains negative

Table 2. Survey weighted regressions of happiness.

	A	B	C
Urban–rural continuum	0.0121***	0.0083**	0.0094**
<i>County-level controls</i>			
Crime rate index		7.198e–06**	7.366e–06*
Persistent poverty		0.0024	0.0144
% Black		–0.0011**	–0.0008+
Low education			0.0174
Housing stress			–0.0013
Low employment			–0.0282
Population loss			–0.0191
Pers. inc. (USD 1000)/cap			0.0002
<i>Person-level controls</i>			
Income	0.0256***	0.0261***	0.0259***
Married or member of an unmarried couple	0.1401***	0.1409***	0.1408***
Unemployed	–0.1683***	–0.1674***	–0.1670***
Age	–0.0077***	–0.0080***	–0.0081***
Age squared	0.0001***	0.0001***	0.0001***
White	–0.0573***	–0.0582***	–0.0561***
Education level	–0.0085+	–0.0111*	–0.0111*
Soc/emo support	0.1846***	0.1841***	0.1843***
General health	0.1342***	0.1357***	0.1356***
N	138,453	132,677	131,657

Note: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

and significant when controlling for a full set of covariates.

Results from all models are consistent. Defining the city in multiple ways and controlling for commonly considered urban problems and other characteristics changes the results little, which suggests that it is the defining features of cities – size and density – that are responsible for the urban–rural happiness gradient, indicating support for hypothesis H2. Solving urban problems, if it were possible, would not likely make people in cities as happy as people are elsewhere. Urbanism – in terms of large population size and high density – is associated with lower happiness.

We use a county fixed effects model to account for potential omitted variable bias, any unobserved heterogeneity, and anything specific about the area that may affect results. US cities differ from each other

substantially. It is not possible to control for all urban problems (or all confounding variables), but any such factors would be picked up in the fixed effects model. Lower happiness in urban areas persists in the fixed effects model.

Finally, we focus on several specific cases and make some illustrative comparisons. A ranking of happy counties (table 8 in the appendix, available online) shows that the happiest counties are not cities, and the least happy counties are large cities. The three happiest counties (happiness > 3.5) in this sample are rural, yet close to large cities. They therefore have the amenities nearby that cities offer, but they escape potential downfalls of larger size and higher density. Douglas County, CO, borders Denver, but remains largely rural and has a population density of 300/sq. mile. Shelby County, TN, is partially urban; it houses Memphis, yet

Table 3. Survey weighted regressions of happiness.

	A	B	C
Population density per sq mile, 05–09 * 1,000,000	–1.5241 +	–0.9695	–1.3860*
<i>County-level controls</i>			
Crime rate index		5.560e–06*	5.825e–06*
Persistent poverty		0.0050	0.0171
% Black		–0.0009*	–0.0006
Low education			0.0165
Housing stress			0.001
Low employment			–0.0101
Population loss			–0.0241*
Pers. inc. (USD 1000)/cap			0.0007
<i>Person-level controls</i>			
Income	0.0254***	0.0258***	0.0255***
Married or member of an unmarried couple	0.1388***	0.1405***	0.1406***
Unemployed	–0.1676***	–0.1671***	–0.1670***
Age	–0.0077***	–0.0080***	–0.0081***
White	–0.0598***	–0.0591***	–0.0559***
Education level	–0.0078	–0.0107*	–0.0109*
Soc/emo support	0.1846***	0.1842***	0.1845***
General health	0.1341***	0.1356***	0.1354***
Age squared	0.0001***	0.0001***	0.0001***
N	138,453	132,677	131,657

Note: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

most of the county is rural and the population density is relatively low at 1200/sq. mile. Johnson County, KS, is mostly rural, yet close to Kansas City and has a population density of 1100/sq. mile. On the other side of the scale, the counties with the lowest happiness scores (happiness < 3.2) in this sample are urban: the first one, St Louis city, MO, is all urban; it covers the central city of St Louis, and has about five times the population density of two of the three happiest counties, at 5700/sq. mile (and 19 times the density of the happiest county). The second and third least happy counties, Bronx, NY and Kings, NY (Brooklyn) are parts of the American city with the largest population, New York City; both counties have a density of over 30,000/sq. mile. The happiness differences between urban and non-urban counties, even though they appear small, are

substantial because most people (>90%) are either satisfied '3' or very satisfied '4' with their lives and the standard deviation across counties in this sample is only 0.06. Hence, the urban–rural happiness gradient is quite substantial relative to the overall variability.

Discussion and future research

Fischer asked the right question about urban unhappiness: is it the city itself – in terms of its core characteristics of size and density – or is it what are typically thought of as urban problems that generate unhappiness? We have excluded alternative explanations (our controls and fixed effects at the county level), that urban problems such as poverty or crime are solely responsible for urban unhappiness. We have confirmed that people

Table 4. Fixed effects survey weighted regressions of happiness.

	FE
Outside centre OR in MSA without centre	−0.003
Inside suburban county of MSA	0.0265
Not in an MSA	0.0927*
Income	0.0261***
Married or member of an unmarried couple	0.1361***
Unemployed	−0.1650***
Age	−0.0076***
Age squared	0.0001***
White	−0.0570***
Education level	−0.0082 +
Soc/emo support	0.1843***
General health	0.1337***
Constant	2.1122***
N	138,453

are less happy in cities, in part because of population size and density, the defining features of the city. Other variables also have an effect, but size and density matter, controlling for urban problems.

There is clearly a ‘city paradox’: cities provide sought-after resources, but we pay a price: the relative unhappiness of city life. High-density living fosters specialisation, information exchange, labour pooling, skills matching, knowledge spillovers, and more. On the other hand it can also foster crime, spread of disease, stress, cognitive and sensory overload, and other problems. Cities act like a magnifying glass, bringing out the best and the worst in us. This synergy has been estimated at the rate of 1.15 relative to the population – doubling population size increases everything (say crime or creativity) that an average person does by 15% (Bettencourt et al., 2007, 2010). Cities do not make people happier as Glaeser (2011) argues. People prefer to live outside of cities (Fuguitt and Brown, 1990; Fuguitt and Zuiches, 1975; Palmer, 2012) and, as we document here, people are happier outside of them.

Yet Americans are moving to metropolitan areas. This may point to an explanation of the Easterlin Paradox (1974) that economic growth does not result in happiness growth; perhaps this is due to people moving to less happy places.⁹ Moving to a city will not necessarily make one less happy, nor will moving to a small town or village make one more happy. How moving to an area affects personal happiness is beyond the scope of this cross-sectional study. Yet, there is an urban–rural happiness gradient and it persists, regardless of urban problems such as poverty or crime. By pointing to urban unhappiness, we do not argue that urbanisation should be countered. There are many benefits to cities, and of course, some people are happier in cities than they would be elsewhere.

This study is about the USA and its results should not be generalised to other countries. At best these results can be generalised to other developed countries. However, in developing countries, there is likely to be a negative relationship between happiness and the city (Berry and Okulicz-Kozaryn, 2009).

As with any study, there needs to be more work done in this area; even though we continue a long-standing line of research, this is the first study to use county-level data to investigate the urban unhappiness hypothesis empirically. Future research may study how neighbourhoods (for example, census tracts) affect the wellbeing of residents. Cities are collections of neighbourhoods, and all characteristics, including happiness and urban problems, are distributed unevenly within a city and across its neighbourhoods. There are notable spatial inequalities (Ballas, 2013), so segregation and inequality are key fields for future work. Finer geographic resolution also has disadvantages; there are no surveys representative of neighbourhoods across a wide range of cities, so a neighbourhood study is not an

alternative or an improvement over the present study but a complement to it. While neighbourhoods often seem independent or self-contained ('a mosaic of little worlds which touch but do not interpenetrate' (Park, 1915: 608)), there are spillovers as well; for example, poverty and crime in one neighbourhood does indirectly affect people and their happiness in other neighbourhoods. While people in public space may not engage in focused interaction, effectively being 'alone, together', people do encounter others in public transportation and in public spaces, especially in large cities (Morrill et al., 2005; Putnam, 2001).

Future research would do well to investigate why people choose to live in cities, given that urbanites are less happy, using person-level panel data. Future work might also explore the effect of the city on the happiness of different social groups. Additional research could use case studies of the happiest cities as opposed to the least happy cities and investigate more deeply than is possible with currently available data what makes residents happy. As more data become available, future research can focus on the exploration of multiple advantages and disadvantages of cities. We have striven to include the most important and easily measurable controls, but it is important to extend this research further by including an even larger set of city characteristics in the future. Notably, segregation, inequality, various types of pollution, and natural amenities are topics for future research in this area (Ballas, 2013; Brereton et al., 2008).

It may be possible that 'deviant' people choose to live in cities – and they may be less happy than are others and so that may depress overall city happiness. Indeed, Wirth (1938) argued that deviance would be higher in cities; Fischer (1995) referred to this as unconventionality, a more positive framing, but with the same notion that people with unusual interests or members of small

subcultures would be more common in cities. These qualities may be to some degree picked up by person-level controls: income, education and unemployment, and by the county-level in the crime rate index. But otherwise, there is room for improvement, and future research could help in this area. In general, though, deviance and 'qualities that make people unhappy' are difficult to measure. There is also a possibility that unhappy people are pulled to cities; people who feel they do not belong in smaller areas and people who are unhappy where they live migrate to cities, but it is possible that even after urban migration their happiness remains. LGBTQ people (e.g. Doderer, 2011) and immigrants (e.g. Damm, 2009) prefer cities. But it is equally possible that happy and energetic extroverts move to cities to realise their full potential.

While there may be an element of self-selection, we think that it is possible that cities have the ability to change people in such a way that they become less happy. For instance, cities may intensify the pecuniary and consumerist orientation in people, and make them more stressed and overworked. Indeed:

To say that self-selection explains some of the differences between town and city is not to say that urbanism is irrelevant to those very same differences. It remains an indirect cause by stimulating selective migration. Although, part of urban/nonurban self-selection is coincidental, most of it results from processes originating in the nature of urbanism itself. (Fischer, 1982: 256)

And this is our interpretation and conclusion: urbanism is at least partially responsible for lower happiness in cities, regardless of urban problems.

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(Okulicz-Kozaryn) on this subject inspired this study – to search for the effects of city size per se as opposed to the effects of compositional differences of bigger cities. We thank Paul Jargowsky and Arielle Kuperberg for their advice and comments, and our fellow affiliated scholars of the Center for Urban Research and Education at Rutgers-Camden for pushing us to be clearer in our argument. We are also grateful to the students in our courses, who have helped us think and rethink ideas about cities, but we take full responsibility for weaknesses in our arguments.

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Notes

1. We use happiness, life satisfaction and (subjective) wellbeing interchangeably, as in much of the literature; happiness is defined in the 'Data and measurement' section. Wirth, who used the term 'malaise' in his influential article, 'Urbanism as a way of life', referred to general dissatisfaction and unhappiness, and hence we use a measure of happiness to study urban malaise, as Fischer (1972, 1973) did. In other words, we interpret the early 20th century focus on 'urban malaise' as 21st century 'urban unhappiness or dissatisfaction'.
2. As a robustness check, we reran our analyses using the regular BRFSS and found similar results, which are available from the corresponding author upon request.
3. The happiness measure used here is quite comprehensive and also popular in the existing literature. This does not preclude the possibility that some aspects of happiness/life satisfaction/wellbeing may be actually improved among city dwellers. This is a topic for future research, but would require a different type of data with a very detailed questionnaire about happiness, or an in-depth qualitative study which would necessarily reduce the size and representativeness of such a data set. For future research, it may be useful to use data sets measuring distinct cognitive and affective aspects of subjective

wellbeing. As of 2015, we are unaware of a data set representative of a large number of US counties that includes such measures.

4. For all categories of both size variables, see Table 7 in the appendix, available online.
5. Only very recently have crime and poverty been suburbanising. This study, however, uses 2005 data.
6. It is better to overcontrol than undercontrol, especially in a large data set as used here (Gujarati, 2002).
7. This survey item does not capture levels of trust or social capital, which may also be important, but it is the best measure available in the BRFSS to measure support, which is related to trust and social capital. Future research should try to use other measures in order to capture trust and social capital.
8. We used the following Stata command: regress <happiness> <person-level variables> <county-level variables> [pw=_cntywt], robust cluster(<county>). We also tried a multilevel logit model, and the results were similar. The bivariate relationship between size of a place and happiness is negative. Stata code and additional results are available upon request from the corresponding author.
9. The Easterlin Paradox shows that Americans did not become happier over the past several decades despite enormous economic progress over that time. The fact that over the same time America (sub)urbanised substantially, and that people are less happy in metropolitan areas as shown here, may help explain the Easterlin Paradox.

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